Riphah School of Computing & Innovation

Updated Detailed FYP Proposal Evaluation Rubrics (15 Criteria)

Category: Research Project (CS, AI, DS, CY) | Total Marks: 100

This updated evaluation framework restructures the Research FYP Proposal evaluation into 15 distinct criteria across 3 major domains: Domain A: Problem Identification & Background Study, Literature & Tool Gap Analysis (50 Marks), Domain B: Research Methodology & Design (25 Marks), and Domain C: Oral Defense, Visual Defense & Technical Mastery (25 Marks). Evaluation in the detailed matrix uses an explicit 1 to 5 Rating Scale (1=Poor, 2=Developing, 3=Satisfactory, 4=Proficient, 5=Exemplary) converted proportionally to assigned marks.

Key Policy & Evaluation Rules:
1. Rating Scale (1–5): Evaluators rate each criterion on a scale of 1 to 5. Final Marks = (Assigned Rating / 5) × Criterion Max Marks.
2. AI Policy Compliance: No AI usage/assistance in core research functionalities or documentation. AI APIs are permitted.
3. Research Focus: Must clearly distinguish original research contributions from simple software/tool implementation wrapping existing literature.

1. Comprehensive Score Breakdown (100 Marks)

#	Domain	Specific Rubric Criteria	Marks
1	**Domain A: Problem Identification & Background Study & Gap Analysis (50 Marks)**	Problem Formulation & Scientific Research Question	5
2		Research Aims, Objectives & Scope Alignment	5
3		Literature & Existing Tool Gap Analysis (Expanded)	10
4		Theoretical & Mathematical Framework	10
5		Dataset Selection, Benchmark & Data Acquisition Protocol	5
6		Expected Final Product / Proof of Concept (PoC) Scope	5
7		Scientific Novelty vs. Adaptation of Existing Research/Tools	5
8		Work Plan, Timeline (WBS/Gantt) & Feasibility (9 hrs/wk)	5
10	**Domain B: Research Methodology Design (25 Marks)**	System Architecture Design	8

11		Methodology and Baseline Benchmarking	9
12		Evaluation Metrics, Validation & Performance Strategy	8
13	**Domain C: Oral Defense & Presentation (25 Marks)**	Presentation Structure, Flowcharts & Visual Diagramming	5
14		Technical Communication, Vocabulary & Professional Delivery	10
15		Q&A Response & Research Methodology Defense	10
	TOTAL SCORE	**Full FYP Proposal Evaluation (15 Criteria Matrix)**	**100**

2. Detailed 15-Criteria Evaluation Matrix (1 to 5 Rating Scale)

Domain A: Written Proposal, Literature & Tool Gap Analysis (50 Marks)

Criteria & Max Marks	Unsatisfactory (0–1):	Developing(2–3)	Proficient / Excellent (4–5):	Marks
1. Problem Formulation & Research Question (5 Marks)	Vague problem statement; fails to clearly state the core research question; fails to demonstrate relevance to CS/AI/DS/CY.	Broad problem definition; research question is stated but lacks scientific precision, clarity, or testable scope.	Precise, well-formulated research question addressing a clear scientific gap with testable hypotheses.	**(5 Marks)**
2. Research Aims & Objectives Alignment (5 Marks)	Objectives are missing, unmeasurable, or describe standard software installation/coding rather than academic research.	Objectives are defined but are generic or partially unmeasurable within an academic project timeline.	Explicit, highly specific, measurable research objectives that directly address the core research question.	**(5 Marks)**
3. Literature & Existing Tool Gap Analysis (10 Marks)	Minimal literature review (<5 sources); relies on blogs/tutorials; fails to compare existing literature or existing software tools.	Summarizes relevant papers chronologically; includes basic tool summary but lacks critical synthesis or clear gap identification.	Comprehensive, critical synthesis of state-of-the-art literature and existing commercial/academic tools explicitly establishing research gaps.	**(10 Marks)**
4. Theoretical & Mathematical Framework	Lacks theoretical foundation; no mathematical	Basic theoretical framework presented but contains minor	Rigorous conceptual, mathematical, or theoretical framework properly	**(10 Marks)**

(10 Marks)	notation, conceptual model, or formal domain logic.	inaccuracies or incomplete mathematical formulations.	grounding the proposed research study.	
5. Dataset Selection & Data Protocol (5 Marks)	No dataset identified; data sources are unclear, inaccessible, invalid, or ethically compromised.	Dataset identified but details regarding volume, features, preprocessing, or splitting strategies are vague.	Clear dataset identification (public benchmark or rigorous primary collection protocol) with clean feature details and partitioning plan.	**(5 Marks)**
6. Expected Final Product / Proof of Concept (PoC) (5 Marks)	Undefined final deliverable; fails to state what proof of concept (PoC), prototype, model, or dataset will be produced.	Vague deliverable scope; mentions a system but lacks specification of the exact proof of concept (PoC) deliverable.	Clear, detailed deliverable specification defining the tangible PoC prototype, API/interface, model checkpoint, and functional artifacts.	**(5 Marks)**
7. Scientific Novelty vs. Adaptation of Existing Tools (5 Marks)	Direct duplication or wrapper of existing tutorial/repository with no novel research contribution or original adaptation.	Incremental novelty; simple application or minor wrapper of an existing tool/research model on a standard dataset.	Clear scientific contribution and novelty (improving accuracy/speed, novel architecture tweak, custom dataset, or novel formulation).	**(5 Marks)**
8. Work Plan, Timeline (WBS/Gantt) & Feasibility (5 Marks)	No work plan or Gantt chart provided; timeline is unrealistic or fails to show required student effort.	Basic timeline provided but lacks clear phase breakdown, WBS task mapping, deliverable mapping, or role breakdown.	Detailed Gantt chart & WBS verifying 9 hrs/week effort per student, clear milestones, phase deliverables, and risk mitigation plan.	**(5 Marks)**

Domain B: Research Methodology & Experimental Pipeline Design (25 Marks)

Criteria & Max Marks	Unsatisfactory (0–1):	Developing(2–3)	Proficient / Excellent (4–5):	Marks
10. System Architecture (8 Marks)	No experimental pipeline or architecture diagram provided; approach is vague and unstructured.	Conceptual diagram present, but data pipeline flows, module interactions, or technical setup details are incomplete.	Clear, detailed experimental pipeline diagram detailing end-to-end data processing, modeling, and evaluation modules.	**(8 Marks)**
11. Methodology, Model Design & Baseline (9 Marks)	No methodology/algorithm specified; fails to define state-of-the-art baselines for comparative evaluation.	methodology/algorithms specified, but baseline selection for comparative performance benchmarking is weak or unjustified.	Well-justified choice of proposed methodology/ algorithms/models alongside explicit, standard baseline models for comparative analysis.	**(9 Marks)**

12. Evaluation Metrics, Validation & Performance (8 Marks)	Metrics missing or inappropriate for the problem (e.g., accuracy for severely imbalanced data).	Standard metrics listed, but validation strategy (k-fold, train/val/test splits, statistical significance) is vague.	Comprehensive evaluation metrics (Precision, Recall, F1, Latency, ROC-AUC) paired with a robust validation plan.	**(8 Marks)**

Domain C: Oral Defense, Presentation & Technical Mastery (25 Marks)

Criteria & Max Marks	Unsatisfactory (0–1):	Developing(2–3)	Proficient / Excellent (4–5):	Marks
13. Presentation Structure, Visuals & Diagrams (5 Marks)	Unstructured, cluttered slides, unreadable text, missing flowcharts or architecture diagrams.	Logical structure but visual aids are standard or overly text-dense in technical slides.	Exceptionally structured slides with clear research flowcharts, high-contrast visual diagrams, and clear typography.	**(5 Marks)**
14. Technical Communication & Vocabulary (10 Marks)	Reads directly from slides, poor time management, weak grasp of domain vocabulary.	Clear voice but relies on visual notes; adequate time management and standard technical vocabulary.	Fluent, professional delivery using precise scientific terms, perfect time management, and balanced team speaking.	**(10 Marks)**
15. Q&A Response & Research Methodology Defense (10 Marks)	Unable to answer technical questions; defensive, evasive, or incorrect domain responses.	Answers surface-level questions but struggles with deeper methodological or baseline justification queries.	Demonstrates deep domain mastery; defends methodology, dataset choices, and theoretical framework logically.	**(10 Marks)**